

PROJECT DOCUMENTATION

Project Name: ServiceNow-Based Employee Laptop Request Application

1. Executive Summary

The ServiceNow-Based Employee Laptop Request Application is designed to streamline and automate the hardware procurement lifecycle for enterprises. By leveraging ServiceNow's native Service Catalog, Catalog Builder, and Flow Designer, this solution eliminates manual email approvals, reduces fulfillment lead times, and ensures complete auditability across IT asset allocations.

1. INTRODUCTION

1.1 Project Overview

The ServiceNow-Based Employee Laptop Request Application is an enterprise IT service management solution developed within the ServiceNow platform. Designed to digitize and automate the lifecycle of IT hardware provisioning, the application provides a centralized Service Catalog interface for employees to request laptops based on their job roles and requirements. By automating manager approval routing, fulfillment task creation for IT support staff, and automated email updates, the application reduces manual administrative overhead and improves time-to-delivery for IT assets.

- 1.2 Purpose
- The primary objective of this project is to eliminate inefficient, paper-based, or manual email-driven hardware provisioning processes. The application aims to:
- Provide a user-friendly, standardized portal for employees to submit laptop requests.
- Automate approval workflows to ensure managerial oversight and policy compliance.
- Streamline IT operations through automated task routing and asset tracking.
- Offer real-time request tracking to improve transparency across departments.

2. IDEATION PHASE

2.1 Problem Statement

- In traditional corporate environments, requesting new or replacement laptops involves fragmented communication channels (emails, physical forms, or unstructured messaging), leading to:
- High request processing delays due to manual approval bottlenecks.
- Miscommunication regarding hardware specifications and availability.
- Lack of real-time visibility for employees regarding request status.
- Inconsistent record-keeping for asset allocation and auditing.

2.2 Empathy Map Canvas

- Says: "I don't know who needs to approve my hardware request." / "It takes weeks to get a replacement laptop when my current one breaks."
- Thinks: "The IT ticketing process is overly complex." / "I hope my request didn't get lost in my manager's inbox."
- Does: Sends follow-up emails manually, contacts IT support repeatedly for status updates, and fills out repetitive details across forms.
- Feels: Frustrated by administrative friction, anxious about delayed onboarding or work downtime due to lack of proper equipment.

2.3 Brainstorming

- Key features identified during solution brainstorming:
- Self-Service Catalog Item: Create a standardized form with dynamic fields (e.g., laptop specifications, justification, department).
- Automated Multi-Stage Workflow: Route requests automatically to line managers and IT fulfillment teams via Flow Designer/Workflows.
- Role-Based Access Control (RBAC): Ensure employees can view only their requests, while managers and IT admins have approval/fulfillment capabilities.
- Notification System: Send automated email notifications on request creation, approval, rejection, and fulfillment.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

- Submission: Employee logs into the Service Portal, selects "Laptop Request" catalog item, inputs requirements, and submits.
- Approval Phase: The system automatically identifies the requester's manager and sends an approval notification.
- Fulfillment Phase: Upon approval, a Service Task (sc_task) is created and assigned to the IT Hardware Asset team.
- Provisioning: IT staff configures the hardware, assigns serial numbers, and updates the task status.
- Closure: Laptop is delivered, the requester confirms receipt, and the request (sc_req_item) is marked as Closed Complete.

3.2 Solution Requirement

- Functional Requirements:
- Catalog Item with mandatory fields (Requester, Laptop Model, Operating System, Business Justification).
- Automated manager approval routing.
- Conditional task generation for IT provisioning upon approval.
- Email notifications for state changes (Requested, Approved, In Progress, Fulfilled, Rejected).
- Non-Functional Requirements:
- Role-Based Security: Restrict record access using Access Control Lists (ACLs) and ServiceNow Roles.

- Performance: Form load time under 2 seconds; response time under 1 second for field dependencies.
- Usability: Intuitive layout conforming to ServiceNow Service Portal design standards.

3.3 Data Flow Diagram

[Employee] ----> (Fill Catalog Form) ----> [sc_request / sc_req_item]

|

v

(Flow Designer / Workflow)

|

+-----+-----+

|

|

(Manager Approved)

(Manager Rejected)

|

|

v

v

[Create sc_task]

[Set Status: Rejected]

|

|

v

v

[IT Fulfillment]

(Trigger Rejection Email)

|

v

[Set Status: Closed]

|

v

(Trigger Delivery Email)

3.4 Technology Stack

- Platform: ServiceNow Personal Developer Instance (PDI) / Enterprise Instance

- Frontend: ServiceNow Service Portal, Catalog Builder, UI Policies, Client Scripts (JavaScript)
- Backend: Flow Designer / Workflow Editor, Business Rules (JavaScript), Tablespace (sc_request, sc_req_item, sc_task)
- Security & Access: Role-Based Access Control (RBAC), Access Control Lists (ACLs)

4. PROJECT DESIGN

4.1 Problem Solution Fit

- The ServiceNow Service Catalog and Flow Designer framework fit the problem requirements by transforming unstructured, manual communications into structured, automated database transactions. Standardized form fields enforce correct input data, while automated workflows eliminate human delay in routing tasks.

4.2 Proposed Solution

- Implement a self-service IT Service Catalog item integrated with ServiceNow's native Workflow/Flow Designer engine.
- Input Stage: Standardized input form with mandatory fields (Requester, Department, Asset Type, Justification).
- Processing Stage: Automated rule-based routing to line managers using system user hierarchy data (sys_user).
- Output Stage: Dynamic task creation (sc_task) for the IT Fulfillment group and automated notification triggers.

4.3 Solution Architecture

- The architecture leverages native ServiceNow table hierarchy:
- Service Portal / Service Catalog Interface: Presentation layer for end-user interaction.

Request Data Layer:

- sys_user: Stores user and manager relationships.
- sc_request: Parent request container.
- sc_req_item (RITM): Stores specific laptop request item details and variables.
- sc_task: Assigned to IT agents for hardware provisioning.

Logic Layer:

- UI Policies / Client Scripts: Frontend validation and dynamic field display.
- Business Rules: Server-side logic for data validation and field updates.
- Flow Designer / Workflow Engine: Manages approvals, notifications, and task lifecycle.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Phase Name	Key Tasks / Deliverables	Estimated/Duration
Phase 1	Requirement Analysis & Planning	Define project scope, create user stories, design empathy map and process flows.	1h 20m
Phase 2	Backend Development & Configuration	Set up custom tables (if needed), catalog variables, UI Policies, and Client Scripts.	2h 00m
Phase 3	Workflow Development & Execution	Build Flow Designer workflows, approval triggers, and task generation rules.	2h 30m
Phase 4	Business Rules, Security & Validation	Configure server-side Business Rules, set up RBAC roles, and apply Access Control Lists (ACLs).	1h 40m
Phase 5	Functional Testing & Deployment	Conduct end-to-end testing, update instance demo links, and finalize project documentation.	1h 30m
Phase 6	Project Presentation & Conclusion	Prepare project presentation and finalize source repository.	1h 00m

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

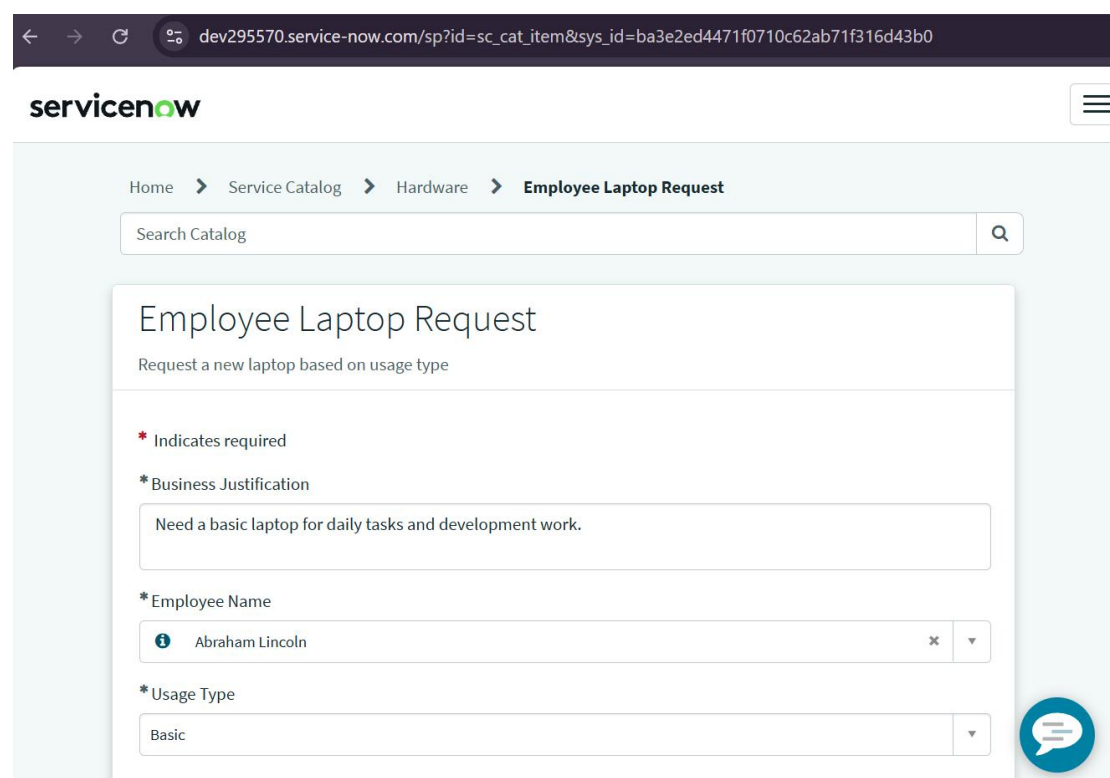
- Functional and performance verification carried out on the instance:
- Catalog Item Validation: Tested mandatory field constraints and dynamic option rendering based on user selection.
- Approval Routing Test: Verified that submitting a request dynamically routes an approval record to the manager under sys_user.manager
- Task Lifecycle Test: Confirmed that approving the request automatically generates an IT fulfillment sc_task and sets the RITM state to Work in Progress.
- Rejection Logic Test: Verified that rejecting the manager approval updates the RITM state to Closed Incomplete.
- Response Time Verification: Checked browser transaction logs to ensure client-side UI Policies execute under 200ms and server-side flow executions finish under 1 second.

7. RESULTS

- 7.1 Output Screenshot

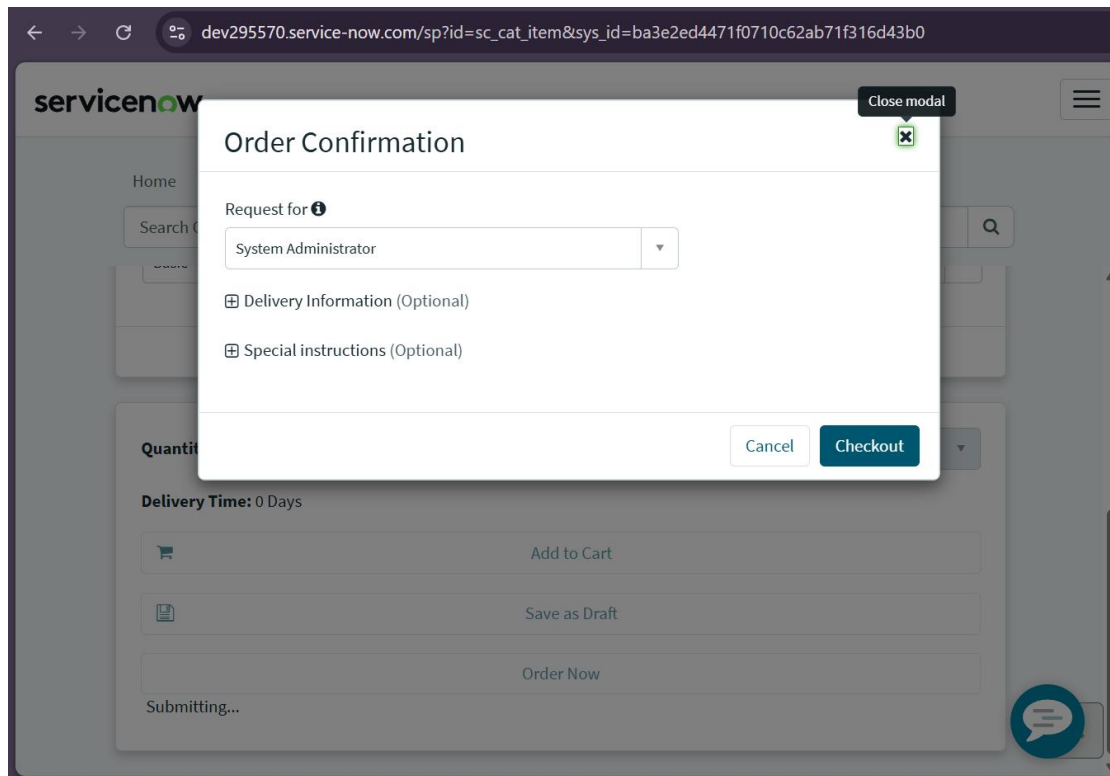
(Insert your 5 saved screenshots into this section)

- Figure 7.1: Employee Laptop Request Catalog Item Form (1_catalog_form.png)

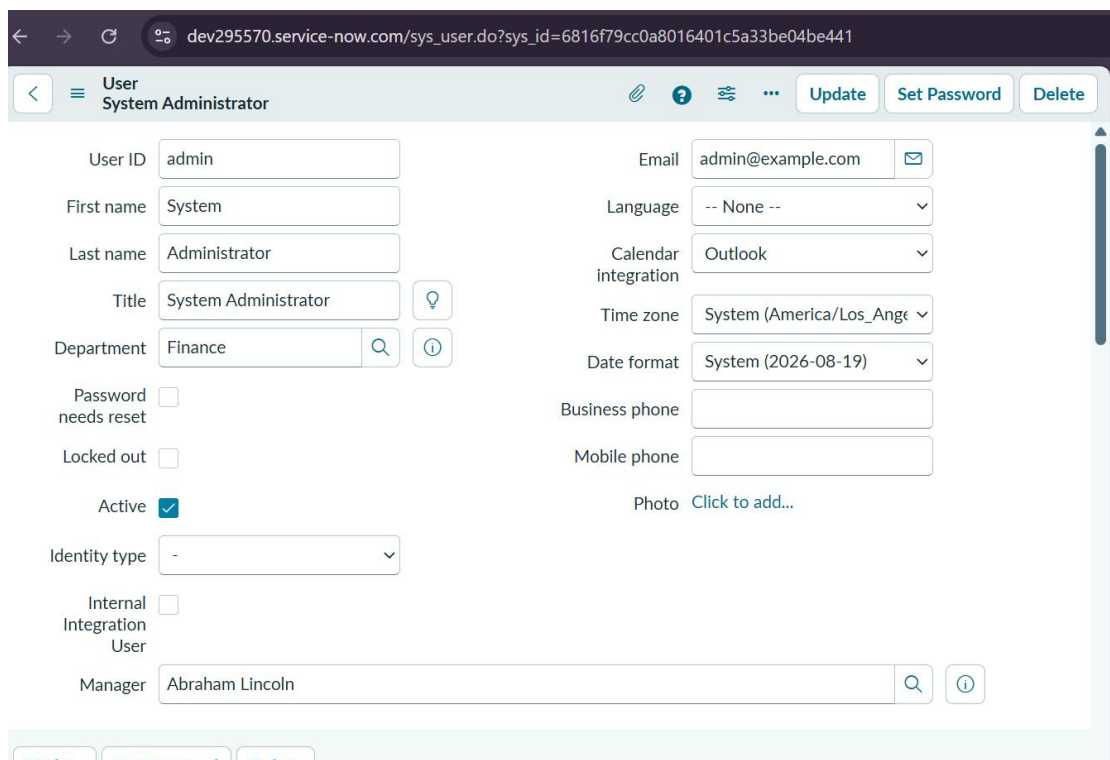


The screenshot shows a web browser window with the URL `dev295570.service-now.com/sp?id=sc_cat_item&sys_id=ba3e2ed4471f0710c62ab71f316d43b0`. The ServiceNow logo is in the top left, and a hamburger menu icon is in the top right. The breadcrumb navigation path is: Home > Service Catalog > Hardware > Employee Laptop Request. Below the breadcrumbs is a search bar labeled "Search Catalog" with a magnifying glass icon. The main form is titled "Employee Laptop Request" with the subtitle "Request a new laptop based on usage type". The form contains several fields: a legend indicating that an asterisk (*) denotes required fields; a "Business Justification" text area with the value "Need a basic laptop for daily tasks and development work."; an "Employee Name" dropdown menu showing "Abraham Lincoln" with an information icon, a close icon (X), and a dropdown arrow; and a "Usage Type" dropdown menu showing "Basic" with a dropdown arrow. A blue chat bubble icon is located in the bottom right corner of the form area.

- Figure 7.2: Manager Approval Notification & Approval Record Page (2_manager_approval.png)



● Figure 7.3: Flow Designer Execution Details / Workflow Visualizer (3_flow_designer.png)



● Figure 7.4: Generated IT Fulfillment Task assigned to Hardware Group (4_it_task.png)

dev295570.service-now.com/sc_req_item.do?sys_id=c8657e8b47ab4bd0c62ab71f316d4334

Requested Item
RTM0010009

Number: RTM0010009
 Item: Employee Laptop Request
 Request: REQ0010009
 Requested for: System Administrator
 Due date: 2026-09-30 02:46:16
 Configuration item:
 Watch list:
 Opened: 2026-09-30 02:46:16
 Opened by: System Administrator
 Stage: Request Approved
 State: Open
 Quantity: 1
 Estimated delivery:
 Backordered:
 Order Guide:
 Variables:
 Business Justification:
 Need a basic laptop for daily tasks and development work.
 Employee Name:
 Abraham Lincoln
 Usage Type:
 Basic
 Comments (Customer visible):
 Comments (Customer visible):
 Work notes:
 Activities: 1
 System Administrator
 Request
 Opened by
 State
 Open
 Related Links
 Catalog Tools
 Approver:
 Search
 Approved for: RTM0010009
 State: Requested
 Approver: System Administrator
 Comments:
 Created: 2026-09-30 02:46:16

● Figure 7.5: End-to-End Status Tracking View (5_ritm_status.png)

dev295570.service-now.com/now/nav/ui/classic/params/target/sc_request.do%3Fsys_id%3D8c657e8b47ab4bd0c62ab71f316d4334

servicenow All : Request - REQ001... Appli... Updat...

Request REQ0010009 Discuss Follow Update Cancel Request Copy Delete

Number: REQ0010009
 Requested for: System Administrator
 Location:
 Due date: 2026-09-30 02:46:16
 Price: \$0.00
 Description:
 Short description:
 Special instructions:
 Comments (Customer visible):
 Comments (Customer visible):
 Opened: 2026-09-30 02:46:16
 Opened by: System Administrator
 Approval: Approved
 Request state: Approved

8. ADVANTAGES & DISADVANTAGES

Advantages

- Increased Efficiency: Automates manual approval routing and task assignments.
- Enhanced Transparency: Real-time request tracking for employees.
- Standardized Data: Enforces mandatory fields to eliminate incomplete requests.
- Scalability: Easily extendable to include additional hardware types or CMDB integration.
- Disadvantages
- Dependency on Accurate System Data: Requires up-to-date user manager relationships (sys_user).
- Platform Overhead: Requires active ServiceNow licensing and developer maintenance.

9. CONCLUSION

- The ServiceNow-Based Employee Laptop Request Application successfully automates and streamlines the hardware provisioning lifecycle. By replacing manual communication with structured Service Catalog forms and automated Flow Designer workflows, the solution improves operational efficiency and ensures transparent asset fulfillment.

10. FUTURE SCOPE

- CMDB Asset Integration: Automatically create and link assets in the Configuration Management Database (CMDB).
- Virtual Agent / Chatbot Integration: Enable laptop requests directly through ServiceNow Virtual Agent or Microsoft Teams.
- Inventory Stock Tracking: Implement automatic stock-level checks before triggering tasks.
- SLA Monitoring: Add Service Level Agreements (SLAs) to track response times.

11. APPENDIX

Source Code / Client Scripts

JavaScript

```
function onChange(control, oldValue, newValue, isLoading) {  
  
    if (isLoading || newValue === '') {  
  
        return;  
  
    }  
}
```

```
if (newValue == 'developer_laptop') {  
  
    g_form.setMandatory('business_justification', true);  
  
    g_form.showFieldMsg('business_justification', 'Developer laptops require explicit  
business justification.', 'info');  
  
} else {  
  
    g_form.setMandatory('business_justification', false);  
  
}  
  
}
```